

Phase 6α / Phase 6β : Closing the Selectivity Gap

Overview of the missing-physics investigation, May 2026

PNPInverse / Forward solver

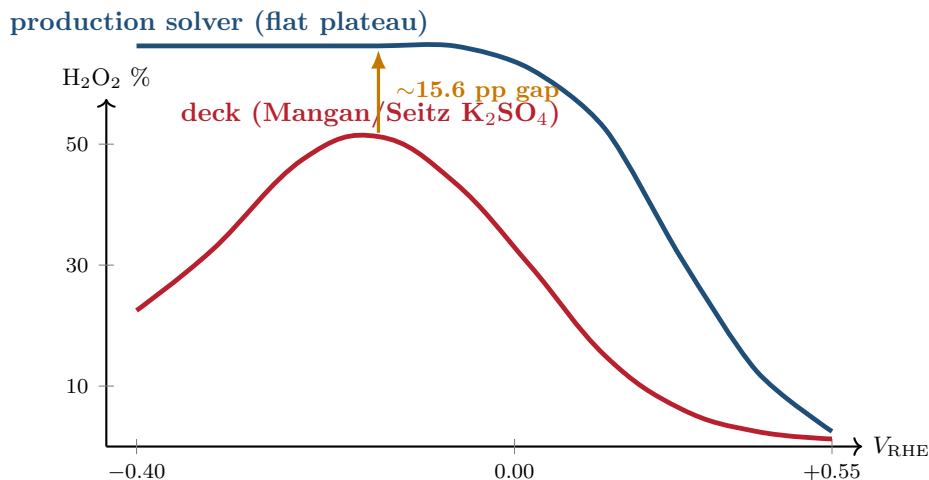
May 13, 2026

Abstract

Light-detail overview of the Phase 6α / Phase 6β work. The production PNP+BV forward stack converges across the deck's voltage band and produces sensible Levich currents, but its H_2O_2 selectivity is flat across V_{RHE} , while the Mangan/Seitz deck data shows a distinct cathodic peak. Phase 6α and 6β are two hypotheses about what missing physics puts that peak there. 6α tested water self-ionization (retired as sub-leading). 6β tested cation hydrolysis at the outer Helmholtz plane following the group's own framing. The 6β architecture and calibration landed cleanly; the fit step (10) hit a non-identifiability verdict and is currently blocked. This document is the short story — see the per-step handoffs in `docs/phase6/` for the long ones.

1 The selectivity gap

The production solver and the deck disagree about one thing in a specific way. Both are at K_2SO_4 , pH 4–6, $L_{\text{eff}} \approx 100 \mu\text{m}$, V_{RHE} in roughly $[-0.4, +0.55]$ V. The solver's total disk current density matches the deck's order of magnitude (Levich-limited cathodic), but the partial H_2O_2 current behaves differently:



The deck shows a clear cathodic peak around $V_{\text{RHE}} \approx 0$ V; the solver produces a flat plateau set by the H^+ diffusion Levich limit ($\approx -0.09 \text{ mA/cm}^2$ at pH 4, $L_{\text{eff}} = 100 \mu\text{m}$). The plateau is real — it's a consequence of the solver's H^+ mass-transport floor — but the peak isn't there in the solver's output. The two phases below are guesses at what missing physics puts the peak there.

2 Phase 6 α — water self-ionization

Hypothesis (6 α): As ORR consumes H^+ near the electrode, water dissociates ($\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$) to backfill surface pH. This raises the available proton flux for the $2e^-$ channel, putting a cathodic H_2O_2 peak where H^+ supply gets boosted by the buffering.

Implementation: proton-condition residual on $E = c_H - c_{OH}$ with closure $c_{OH} = K_w^{\text{eff}}/c_H$. Wired default-off (`enable_water_ionization=False`) so the baseline is byte-equivalent to pre-6 α ; opt-in via a K_w^{eff} outer ladder on `solve_anchor_with_continuation`.

The Phase 6 α L_{eff} transport sweep (8 boundary thicknesses $\times$ 13 voltage points, ~ 58 min) converged cleanly. Three falsification gates:

P1 PASS — Levich linearity preserved across L_{eff} .

P2 PASS — no spurious cathodic peak introduced.

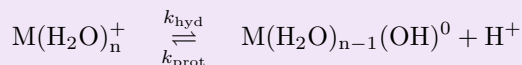
P3 FAIL — max surface pH was 10.58 at $L_{\text{eff}} = 16 \mu\text{m}$, and the surface pH was *invariant* with L_{eff} . A textbook bulk water-dissociation rate estimate ($\sim 1.4 \text{ mol/m}^3/\text{s}$) is $\sim 20\times$ slower than the cathodic H^+ demand the model needs to sustain.

Water does autoionize, but it's **sub-leading** as a buffer at deck conditions. Phase 6 α retired as the primary mechanism; plumbing preserved as opt-in.

3 Phase 6 β — cation hydrolysis at the OHP

The pivot followed the group's own published framing (Linsey 2025 ACS-CATL deck, Co-Zhang 2019 *Angewandte*, Singh et al. 2016 *JACS*). The proposed mechanism:

Cation hydrolysis at the outer Helmholtz plane:



A water molecule in the cation's hydration shell deprotonates near a polarized cathode.

The cation's effective $\text{p}K_a$ shifts sharply with the local Stern field (Singh 2016 Eq. 4):

$$\log_{10}(K_a^{\text{eff}}/K_a^{\text{bulk}}) \propto e \sigma_S r_M / (k_B T \epsilon_{\text{OHP}} \epsilon_0).$$

Bulk K^+ $\text{p}K_a \approx 14.5$; near a polarized Cu cathode it drops to ~ 8.5 . Cs^+ drops further to ~ 4.3 , which matches the cation-dependence in the deck.

The mechanism is attractive because it's cation-specific: K^+ , Cs^+ , Na^+ , Li^+ each have their own bulk $\text{p}K_a$ and Singh shift, and the deck's cation series should fall out from one set of equations.

4 Architecture (v9) and sub-step chronology

The v9 architecture (approved after a five-round GPT critique) made these choices:

- **Dynamic K^+ .** K^+ becomes a fourth Nernst–Planck species (4-DOF stack: O_2 , H_2O_2 , H^+ , K^+). No analytic Boltzmann counterion for K^+ — the hydrolysis reaction couples through K^+ ’s spatial profile, which has to be a real DOF.
- **Global Γ_{MOH} on the electrode.** Surface MOH coverage is a single scalar Real-element function on the electrode boundary (mol/m^2), evolved by finite-rate kinetics.
- **Finite-rate kinetics + Langmuir cap.** $R_{net} = k_{hyd} c_{M^+}(0) (1 - \Gamma/\Gamma_{max}) - k_{prot} c_H(0) \Gamma/\delta_{OHP}$; desorption removal $R_{des} = k_{des} \Gamma$ treated as an open reservoir.
- **No Stern surface-charge correction.** Cation hydrolysis couples to Poisson purely through dynamic c_{K^+} ; no separate surface-charge bookkeeping.

Sub-steps after v9, in chronological order:

Step	What changed	Outcome
v10a	Langmuir cap $(1 - \Gamma/\Gamma_{max})$ on the hydrolysis rate; $\Gamma_{max} \approx 0.047$ nondim ($\approx$ one monolayer of hydrated MOH at the OHP).	Closed-form Picard converges; saturation behaves smoothly.
v10a’ V-sweep	$C_S = 0.20$ F/m ² (Bohra-Koper-Choi consensus) and K_0^{R4e} factor = 10^{-14} to probe $2e^-/4e^-$ coexistence; V-sweep across the deck band.	$V_{kin} = -0.10$ V passes the primary route ($\sigma_S < 0$, branch active, not transport-artifact, not cap-dominated); K^+ enrichment dominates F_0 cathodically.
A.2 k_{hyd} grid	10-rung k_{hyd} ladder at V_{kin} , $\lambda=1.0$.	10/10 rungs converge, mass-balance at machine precision; selectivity gap at V_{kin} is only +5 pp, so Γ_{max} / k_{des} retune flagged <i>low priority</i> .
Step 6 plumbing	Four wiring paths verified (form-build residual, pK_a -shift context, Picard Γ update, diagnostics) via ablation.	All pass; physics finding: K^+ Boltzmann pile-up ($c_K \approx 291$ nondim at OHP) makes a 5% sentinel floor unreachable below $R_{inj} \leq 10$.
v10b calibration	Literature-grounded lock of three parameters: $\Gamma_{max}=0.047$, $k_{des}=1.0$ (Eyring prior $\Delta G \approx 0.80$ eV), $C_S=0.20$ F/m ² . Sensitivity sweeps D7-D1 (C_S bracket) and D7-D4 ($3 \times 5 \times 2$ matrix).	All hard gates pass ($cd < 0$, R_{4e} sign preserved, $R_{net} \geq 0$, mass-balance rel $\leq 5 \times 10^{-3}$).
Step 9 B.2	Densified $14 \times 10 = 140$ -rung $k_{hyd} \times \lambda$ grid at V_{kin} .	14/14 k_{hyd} converge at $\lambda=1$ (after step 9.5 added warm.start.floor bisection for $\lambda=0$ rungs); mass-balance $\leq 5 \times 10^{-13}$.
Step 10 Phase D	Fit a single scalar Δ_β (carbon-vs-Cu offset in Singh Eq. 4) against the deck K_2SO_4 H_2O_2 selectivity at pH 3.5–4.5 (deck mean 50.95 pp). Two paths tested: Stern σ -mapping and ablation σ -override.	OUTCOME.C_NON_IDENTIFIABLE flagged — Δ_β alone cannot match the deck. See callout below.

5 Current state — the Phase D verdict

Step 10 Phase D verdict (2026-05-12): OUTCOME_C_NON_IDENTIFIABLE flagged

Δ_β is not the right knob for closing the selectivity gap at the v10b production point. Two facts force the verdict:

(i) **Stern σ -mapping is degenerate.** Loss is flat at 15.629 pp² across *11 orders of magnitude* of Δ_β ($-1.55 \times 10^7 \rightarrow 0$ pm²). The local Stern σ at the OHP is $\sim 10^{-7}$ counts/pm², so rescaling Δ_β barely moves the Singh shift in this regime.

(ii) **Uniform +15.6 pp overshoot.** Across all Δ_β tested, the model's max H₂O₂% is 66.58 pp. Deck mean is 50.95 pp. Δ_β cannot close 15.6 pp of overshoot. This is the primary acceptance-gate failure.

Phase D plan Risk #4 (*non-identifiable under local Stern σ*) is confirmed. Phase E must not launch on this Δ_β .

6 Open issues and what's blocked

1. **Steric coefficient discrepancy for dynamic species** (surfaced 2026-05-12 during Phase D bridge diagnostics). O₂, H₂O₂, H⁺ are all seeded with A_DEFAULT=0.01, which corresponds to a hard-sphere radius $r \approx 14.9$ Å — $\sim 150\times$ larger than the physical values (O₂ $r=1.7$ Å, H₂O₂ $r=2.0$ Å, H⁺ $r=2.8$ Å for H₃O⁺ Stokes). H⁺ in particular is therefore clamped $\sim 150\times$ tighter than its real Bikerman cap would allow. The Phase D *non-identifiability* verdict is independent of this, but the +15.6 pp overshoot might not be — *anything that depends on the Levich plateau setting H⁺ supply is suspect until the physical-a bridge runs land*.
2. **Δ_β cannot close the gap alone.** Phase D' / Phase 6 γ candidate scopes (in rough order of likelihood):
 - Re-fit k_{des} and Γ_{max} (Phase D locked these as v10b-calibrated; falsification re-opens them).
 - $r_{\text{H,El}}$ sensitivity sweep — the Cu prior (200.98 pm) may not transfer cleanly to CMK-3 carbon.
 - Local-pH / mass-transport coupling — model selectivity is V-flat, which suggests it's transport-limited and the H⁺ Levich convention may need re-examination.
 - Singh formula *structure* (Plan §3.1 locked the form as a hard invariant; if Δ_β -alone fails the structure may itself need revisit — this is a bigger paradigm change).
3. **Phase 6 δ (parallel alkaline ORR kinetics) likely still needed.** Cation hydrolysis fixes the local-pH *source*, not the downstream decay mechanism. If 6 β only ever produces a plateau, 6 δ .1 (explicit $R_{2\text{e,alk}} + R_{4\text{e,alk}}$ parallel reactions) would be the next step — the order is then 6 β .1 $\rightarrow$ measure $\rightarrow$ 6 δ .1 $\rightarrow$ measure.
4. **Singh 2016 SI extraction.** Phase D used a placeholder $\Delta pK_a = \beta_M \text{sgn}(\sigma_S) |\sigma_S|^p$ with $p \in \{0.5, 1.0, 2.0\}$. A clean re-read of the Singh SI for the exact functional form (sign, ln 10 factor, distance dependence) is queued.
5. **Cation-series transferability (Phase E).** β values from Singh 2016 are CO₂RR-on-Cu; transferability to ORR-on-carbon is an assumption. Phase E is a *predictive screen*, not a decisive

falsifier — if the (Cs, Na, Li) holdout fails, audit the Stern/OHP field mapping before rejecting the cation-hydrolysis hypothesis outright.

Net summary

What landed cleanly: the architecture (v9 dynamic $K^+ + \Gamma_{\text{MOH}} + \text{finite-rate kinetics}$), the literature calibration (v10b $\Gamma_{\text{max}}, k_{\text{des}}, C_S$), and all the plumbing (step 6, step 9, step 9.5 floor bisection). All convergence and mass-balance gates pass.

What didn't land: the *fit*. Δ_β degeneracy at the production Stern σ means the parameter Plan D was built around can't pull selectivity down by the needed ~ 15.6 pp.

What's holding things up: (a) the steric-coefficient bridge runs disambiguating whether the H^+ Bikerman clamp is silently producing the overshoot, and (b) a decision on whether to open $k_{\text{des}}/\Gamma_{\text{max}}$ as additional fit parameters or revisit the Singh formula structure itself. Until (a) and (b) resolve, Phase E remains blocked.

See also:

- docs/phase6/PHASE_6A_INVESTIGATION_SUMMARY.md — Phase 6α outcome + Phase 6β scoping handoff.
- docs/phase6/phase6b_next_steps_plan.md — live Phase 6β plan.
- docs/phase6/phase6b_step10_phase_D_summary.md — Step 10 Phase D detailed writeup.
- docs/phase6/v10b_calibration_summary.md — v10b $\Gamma_{\text{max}} / k_{\text{des}} / C_S$ literature grounding.
- docs/phase6/CONJECTURE_AUDIT_2026-05-09.md — Cs^+ vs deck-baseline K^+ audit.
- Companion writeup *Forward Solver Codepath* ([forward_codepath_demo_slide15.pdf](#), this directory) — what the solver *does* converge cleanly when $6\alpha / 6\beta$ are off (default-off byte-equivalent baseline).